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Reference Material on Mathematics-I (MTH1001)

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Mathematics-I (MTH1001)

Unit-IV

Vector Integral Calculus:

Green's theorem, Surface integrals, Volume integrals, Gauss Divergence theorem and Stokes theorem.

MODULE-4

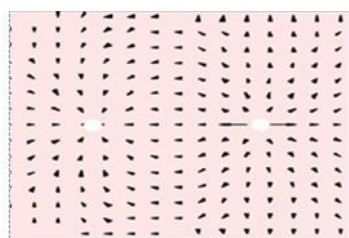
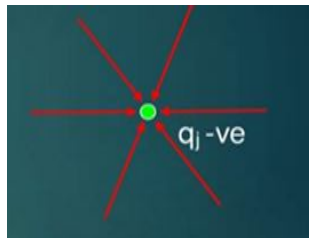
4.1. Concepts covered

- Application of Gauss divergence theorem
- Divergence theorem of Gauss
- Physical significance of Gauss divergence theorem.
- Examples of Gauss divergence theorem
- Home work

4.1.1 Application of Gauss divergence theorem.

Electric field

- Field lines gives a local direction of field
- Field around positive charge directed away from the charge
- Field around negative charge directed towards the charge
- Principles of superposition use for field due to a diopole(+ve and –ve charge combination)



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4.1.2 Divergence theorem of Gauss

(Translation between volume and surface integral)

Again Let T be a closed bounded region in space whose boundary is piecewise smooth surface. Let $F(x, y, z)$ be a vector function that is continuous first partial derivatives in some domain containing T. Then

$$\iiint_T \operatorname{div} F = \iint_S F \cdot n \, dA \quad (1)$$

Where n is the outer unit normal vector.

$$\iiint_T \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dx \, dy \, dz = \iint_S (F_1 \cos \alpha + F_2 \cos \beta + F_3 \cos \gamma) dA \quad (3)$$

Where $n = (\cos \alpha, \cos \beta, \cos \gamma)$

Also we have expressed as

$$\iiint_T \left(\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \right) dx \, dy \, dz = \iint_S (F_1 \, dy \, dz + F_2 \, dz \, dx + F_3 \, dx \, dy) \quad (4)$$

4.1.3 Physical significance of Gauss divergence theorem.

- The divergence of a vector field can be positive negative or zero.
- If the divergence of a vector field at any point is positive, it means that flux entering through the surface is less than the flux coming out through that surface.
- If the divergence of a vector field at any point is negative, it means that flux entering through the surface is more than the flux coming out through that surface.
- If the divergence of a vector field at any point is zero, it means that flux entering through the surface is equal to the flux coming out through that surface.

Examples of Gauss divergence theorem

Example.1 Evaluate $\iint_S (x^3 \, dy \, dz + x^2 y \, dz \, dx + x^2 z \, dx \, dy)$

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Where S is the closed surface consisting of the cylinder $x^2 + y^2 = a^2$ ($0 \leq z \leq b$) and the circular disks $z = 0$ and $z = b$ ($x^2 + y^2 \leq a^2$).

Solution

Here

$$F_1 = x^3, F_2 = x^2 y, F_3 = x^2 z$$

$$\text{Hence } \operatorname{div} F = 5x^2$$

Introducing the polar coordinate r, θ by $x = r \cos \theta, y = r \sin \theta, dx dy dz = r dr d\theta dz$, then we obtain

$$\iiint_T 5x^2 dx dy dz = 5 \int_{z=0}^b \int_{r=0}^a \int_{\theta=0}^{2\pi} r^2 \cos^2 \theta r dr d\theta dz = \frac{5}{4} \pi b a^4.$$

Assignment.

Evaluate by divergence theorem $\iint_S F \cdot n dA$

1. $F = [e^x, e^y, e^z]$ S the surface of the cube $|x| \leq 1, |y| \leq 1, |z| \leq 1$.

2. $F = [x^2, y^2, z^2]$ S the sphere $x^2 + y^2 + z^2 = 9$.

3. $F = [x, y, z]$ S the sphere $x^2 + y^2 + z^2 = 4, z \geq 0$.

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: Stokes's theorem and applications

: Application of Stokes' theorem

Objectives

In this section you will learn the following:

- Computational applications of Stokes' theorem.
- Physical applications of Stokes' theorem.
- Sufficient conditions for a vector field to be conservative.

4.2.1 Applications of Stokes' theorem

Stokes' theorem gives a relation between line integrals and surface integrals. Depending upon the convenience, one Integral can be computed in terms of the other.

Example.1 (computation of line integral):

We want to compute

$$\oint_C \mathbf{F} \cdot d\mathbf{r}, \text{ where } \mathbf{F} = y\mathbf{i} + xz^3\mathbf{j} - zy^3\mathbf{k}$$

and C is the circle in the plane $z = -3, x^2 + y^2 = 4$, oriented anti-clockwise. To apply Stokes' theorem, let us find a convenient S surface $C: x^2 + y^2 \leq 4, z = -3$. $\partial(S) = C$ whose boundary is

The most natural surface in this case is the circular disc

For S if we choose the normal vector $\mathbf{n}$ to be then will have anti-clockwise orientation, and by Stokes' s theorem we will have

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS$$

Since,

$$\text{curl}(\mathbf{F}) \cdot \mathbf{n} = \left[\frac{\partial}{\partial x} (xz^3) - \frac{\partial}{\partial y} (y) \right] = z^3 - 1.$$

we have

$$\oint_C (\mathbf{F} \cdot d\mathbf{r}) = \iint_S (z^3 - 1) dS = \iint_S (-28) dS = -28(4\pi).$$

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Example:2

Let us try to apply the above technique to evaluate

$$\oint_C (\mathbf{F} \cdot d\mathbf{r}) \text{ where } \mathbf{F} = -\frac{y}{x^2+y^2} \mathbf{i} + \frac{x}{x^2+y^2} \mathbf{j},$$

and C is the circle $x^2 + y^2 = 1, z = 0$ oriented clockwise. We note that for $\mathbf{F}$, $\text{curl}(\mathbf{F}) = 0$, and hence

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} dS = 0,$$

Whatever be the surface S , as long as $\partial(S) = C$. We also know (see example 47.2.8) that

$$\oint_C (\mathbf{F} \cdot d\mathbf{r}) = -2\pi.$$

Thus,

$$\oint_C \mathbf{F} \cdot d\mathbf{r} \neq \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS.$$

Does this contradict Stokes' theorem? The answer is no. The reason for this is that Stokes' theorem is not applicable for the given $\mathbf{F}$ because

$$\mathbf{F}(x, y, z) = -\frac{y}{x^2+y^2} \mathbf{i} + \frac{x}{x^2+y^2} \mathbf{j}$$

is defined in $D := \{(x, y, z) \in \mathbb{R}^3 \mid (0, 0, z)\}$. Thus, there does not exist any surface $S \subseteq D$ Whose boundary is the unit circle $C: x^2 + y^2 = 1$ in the xy -plane. However, we can still evaluate the integral

$$\oint_C \mathbf{F} \cdot d\mathbf{r}$$

as follows: Consider the vector field

$$\tilde{\mathbf{F}}(x, y, z) := -\frac{y}{x^2+y^2+z^2} \mathbf{i} + \frac{x}{x^2+y^2+z^2} \mathbf{j},$$

$(x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$. Note That $\tilde{\mathbf{F}} = \mathbf{F}$ on and if we consider the upper hemisphere S given by $x^2 + y^2 + z^2 = 1, z \geq 0$, then $\partial(S) = C$ and $S \cup C \mathbb{R}^3 \setminus \{(0, 0, 0)\}$. If we select the outward unit normal

$$\mathbf{n} = x \mathbf{i} + y \mathbf{j} + z \mathbf{k},$$

$x^2 + y^2 + z^2 = 1$ on S , then $\partial(S) = C$ gets the anti-clockwise orientation and we have by Stokes' theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C \tilde{\mathbf{F}} \cdot d\mathbf{r} = \iint_S (\text{curl } \tilde{\mathbf{F}}) \cdot \mathbf{n} dS.$$

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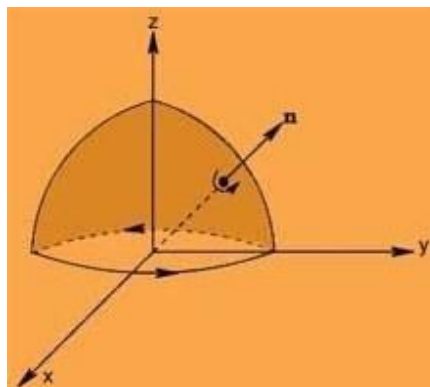


Figure: Orientation $\mathbf{n}$ and

Let S be given the parameterizations :

$$\mathbf{r}(x, y) = x\mathbf{i} + y\mathbf{j} + (\sqrt{1 - x^2 - y^2})\mathbf{k}, (x, y) \in D.$$

where D is the unit disc in xy -plane. Then

$$\mathbf{r}_x \times \mathbf{r}_y = -f_x \mathbf{i} - f_y \mathbf{j} + \mathbf{k},$$

for $f(x, y) = \sqrt{1 - x^2 - y^2}$. This gives

$$\mathbf{r}_x \times \mathbf{r}_y = +\frac{x}{z}\mathbf{i} + \frac{y}{z}\mathbf{k} + \mathbf{k}.$$

Also

$$\text{curl}(\mathbf{F}) = -2zx\mathbf{i} - 2zy\mathbf{k} - z^2\mathbf{k}.$$

Hence

$$\begin{aligned} \iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) \, ds &= \iint_D \left(+\frac{x}{z}\mathbf{i} + \frac{y}{z}\mathbf{k} + \mathbf{k} \right) \cdot (-2zx\mathbf{i} - 2zy\mathbf{k} - z^2\mathbf{k}) \, dxdy \\ &= \iint_D -(2x^2 + 2y^2 + 2z^2) \, dxdy \\ &= -2 \iint_D dxdy \\ &= -2\pi. \end{aligned}$$

4.2.2 Example (Calculation of surface integral):

In the formula

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$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl}(\mathbf{F}) \cdot \mathbf{n} dS$$

the only relation between C and S is that $\partial(S) = C$. Thus, if S_1 and S_2 are two surfaces such that $\partial(S_1) = \partial(S_2) = C$

and $\mathbf{F}$ is defined in a region including have
$$\iint_{S_1} \text{curl}(\mathbf{F}) \cdot \mathbf{n} dS = \oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_{S_2} \text{curl}(\mathbf{F}) \cdot \mathbf{n} dS$$

This is useful in computations. We consider an example. Let us evaluate

$$\iint_S (\nabla u \times \nabla v) \cdot \mathbf{n} dS$$

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where

$$u = x^3 - y^3 + z^3, \text{ and } v = x + y + z,$$

S is the upper hemispherical sheet

$$x^2 + y^2 + z^2 = 1, z \geq 0,$$

$\mathbf{n}$ is the unit normal to S with non-negative z -component. Since

$$(\nabla u \times \nabla v) = \text{curl}(u \nabla v),$$

we have by Stokes' theorem

$$\begin{aligned} \iint_S (\nabla u \times \nabla v) \cdot \mathbf{n} \, dS &= \oint_C (u \nabla v) \cdot d\mathbf{r} \\ &= \iint_{S_1} (\nabla u \times \nabla v) \cdot \mathbf{n} \, dS \end{aligned}$$

where we select S_1 be the circular disc $D = \{(x, y, z) \mid x^2 + y^2 = a^2, z = 0\}$.

Thus

$$\iint_S (\nabla u \times \nabla v) \cdot \mathbf{n} \, dS = \iint_D (\nabla u \times \nabla v) \cdot \mathbf{k} \, dx dy.$$

Since, $(\nabla u \times \nabla v) \cdot \mathbf{k} = 3x^2 + 3y^2$, we have

$$\begin{aligned} \iint_C (\nabla u \times \nabla v) \cdot \mathbf{n} \, dS &= \iint_{x^2+y^2 \leq a^2} 3(x^2 + y^2) \, dx dy \\ &= 3 \int_0^a \int_0^{2\pi} r^3 \, dx dy \\ &= \frac{3\pi}{2}. \end{aligned}$$

4.2.3 Note (Green's theorem, a particular case of Stokes' theorem):

Consider a planer vector-field.

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}.$$

Let D be a region in the xy -plane with $\partial(D) = C$ a simple closed curve. If we treat D as a flat surface, oriented $\mathbf{k}$ as the unit normal, then by Stokes' theorem, treating $\mathbf{F}$ as a vector field in z -space,

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j} + 0\mathbf{k}$$

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D \text{curl}(\mathbf{F}) \cdot \mathbf{k} \, dx dy$$

which is Green's theorem, since

$$\text{curl}(\mathbf{F}) = \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}.$$

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We showed in section 47.2, that given points $A, B \in D \subseteq \mathbb{R}^3$ and a continuously differentiable vector field on D if

$$\int_C \mathbf{F} \cdot d\mathbf{r}$$

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is independent of the path C going A to B , then $\text{curl}(\mathbf{F}) = 0$. We also stated that this condition is also sufficient if the domain D is simply connected. We prove this as an application of Stokes' theorem.

4.2.4 Theorem (Conditions for conservativeness):

Let D be a simply connected domain in $\mathbb{R}^3$ $\mathbf{F}: D \rightarrow \mathbb{R}^3$ be a continuously differentiable vector field. Then the following are equivalent:

1. $\mathbf{F}$ is conservative.
2. $\int_{C(A,B)} \mathbf{F} \cdot d\mathbf{r}$ is independent of the path $C(A,B)$ joining A to B .
3. $\text{curl}(\mathbf{F}) = 0$.

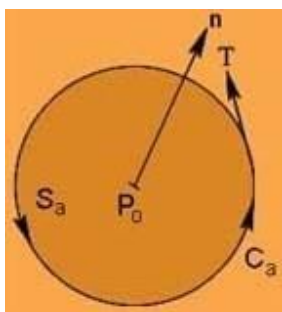
In view of the statements above, we only have to show that $(iii) \Rightarrow (i)$. Let C be any simple closed curve in D . Since D is simply connected, we can find an Orientable piecewise smooth surface $S \subseteq D$ such that $C = \partial S$. Then by Stokes' theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl}(\mathbf{F}) \cdot \mathbf{n} d\mathbf{r} = 0.$$

4.2.5 Physical interpretation of Curl:

Stokes' theorem provides a way of interpreting the curl of a vector-field $\mathbf{F}$ in the context of fluid-flows. Consider a small circular disc S_a of radius a at a point P in the domain of $\mathbf{F}$. Let $\mathbf{n}$ be the unit normal to the disc at P_0 . Then by Stokes' theorem

$$\begin{aligned} \oint_{C_a} (\mathbf{F} \cdot \mathbf{T}) ds &= \iint_{S_a} \text{curl}(\mathbf{F}) \cdot \mathbf{n} d\mathbf{r} \\ &= \iint_{S_a} (\text{curl}(\mathbf{F}) \cdot \mathbf{n}) d\mathbf{r} \\ &\approx (\text{curl}(\mathbf{F})(P_0) \cdot \mathbf{n}(P_0)) \iint_{S_a} dS \\ &= (\text{curl}(\mathbf{F})(P_0) \cdot \mathbf{n}) A(S_a) \end{aligned}$$



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Figure: Flux along C_2

Thus,

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$$(\text{curl } \mathbf{F})(P_0) \cdot \mathbf{n}(P_0) \cong \frac{1}{A(S_\alpha)} \oint_{C_\alpha} (\mathbf{F} \cdot \mathbf{T}) ds$$

Note that $\mathbf{F} \cdot \mathbf{T}$ gives the component of $\mathbf{F}$ in the direction of the tangent and hence gives the rotational component of $\mathbf{F}$ along ds . Then $\oint_{C_\alpha} (\mathbf{F} \cdot \mathbf{T}) ds$ is called the **circulation density** of $\mathbf{F}$ around C_α . If we let $\alpha \rightarrow 0$ in (92), then we will have (as error in approximation goes to zero)

$$(\text{curl } \mathbf{F})(P_0) \cdot \mathbf{n}(P_0) = \lim_{\alpha \rightarrow 0} \frac{1}{A(S_\alpha)} \oint_{C_\alpha} (\mathbf{F} \cdot \mathbf{T}) ds$$

For this reason, the normal component of $\text{curl}(\mathbf{F})$, also called the **specific circulation** of the fluid at the point P_0 . Note that the specific circulation is maximum when $\text{curl}(\mathbf{F})(P_0)$ and $\mathbf{n}(P_0)$ have the same direction we can interpret Stokes' theorem.

$$\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{n}) dS = \oint_C \mathbf{F} \cdot d\mathbf{r}$$

as follows: the collective measure of rotational tendency is equal to the tendency of the fluid to circulate around its boundary. Thus, if $\text{curl } \mathbf{F} = 0$ in S , then

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$$

i.e., there is no circulation tendency, or one says the fluid is **irrotational**.

Practice Exercises

- Using Stokes' theorem, evaluate the integral

$$\int_C \mathbf{F} \cdot d\mathbf{r},$$

where

- $\mathbf{F}(x, y, z) = 2y\mathbf{i} + 3z\mathbf{j} + x\mathbf{k}$, and C is the triangle with vertices $A(0, 0, 0), B(0, 2, 0), C(1, 1, 1)$ oriented for A to B to C to A .
- $\mathbf{F}(x, y, z) = z^2\mathbf{i} + y\mathbf{j} + xz\mathbf{k}$, and C is the boundary of the upper hemi-sphere $S: z = \sqrt{4 - x^2 - y^2}$, oriented counter clockwise
- $\mathbf{F}(x, y, z) = x^2\mathbf{i} + 4xy^3\mathbf{j} + y^2x\mathbf{k}$ and C is the rectangle going $A(0, 0, 0), B(1, 0, 0), C(1, 3, 2), D(0, 3, 2)$, oriented from A to B to C to D to A .
- $\mathbf{F}(x, y, z) = 3y\mathbf{i} + 4x\mathbf{j} + 2y\mathbf{k}$, and C is the boundary of the paraboloid $z = 4 - x^2 - y^2, z \geq 0$ oriented counter Clock wise

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